🧩 M002 Extension 1 — Redstone Trail Solver

Topic: Redstone Trail Solver (3D Cube Maze) · **Course:** Maze Madness · **Level:** Extension (Advanced) · **Time:** about 60 minutes

The agent solves a **3D cube maze** on its own, reading a **redstone trail** all around it to decide each step. Drive it with chat: `l` turns left, `r` turns right, `run` starts the solver, `rl` teleports it back.

1 · Read the Chat Commands 🎛️

Each block binds a **chat word** to an action. Type the word, the agent runs the code.

```
def turn_left():
    agent.turn(LEFT_TURN)
player.on_chat("l", turn_left)
```

What does the agent do when you type `l`?

```
def go_back():
    agent.teleport_to_player()
player.on_chat("rl", go_back)
```

You type `rl`. Where does the agent go? When is that useful?

Why bind small helpers (`l`, `r`, `rl`) *before* you write the big `run` solver?

2 · Trace the Solver 🔍

This loop runs when you type `run`. It checks redstone in different directions and reacts.

```
while True:
    agent.move(FORWARD, 1)
    if agent.detect(REDSTONE, DOWN) and agent.detect(REDSTONE, RIGHT):
        agent.move(FORWARD, 1)
        agent.interact(LEFT)
    elif agent.detect(BLOCK, RIGHT) and agent.detect(REDSTONE, LEFT):
        agent.move(FORWARD, 1)
        agent.turn(LEFT_TURN)
        agent.move(FORWARD, 1)
    elif agent.detect(REDSTONE, DOWN):
        agent.move(UP, 1)
    elif agent.detect(REDSTONE, UP):
        agent.move(DOWN, 4)
```

`while True` keeps going. When does it stop on its own?

What does `detect(REDSTONE, DOWN)` tell the agent?

The maze is 3D. Which two lines move the agent between floors? What does each do?

Why does the order of the `if` / `elif` checks matter? What changes if the `elif agent.detect(REDSTONE, DOWN)` check came *first*?

3 · Spot the Bug 🐛

Read what each block should do, rewrite it, then explain the fix.

Bug A — The solver should keep going until the maze is done, checking the trail every step.

```
agent.move(FORWARD, 1)
if agent.detect(REDSTONE, DOWN):
    agent.move(UP, 1)
```

Hint: this checks only once. What wraps a block so it repeats?

Write the fix:

Why was it wrong? Why does your fix work?

Bug B — When there is redstone **down and right**, step forward, then flip a lever on the **left**.

```
if agent.detect(REDSTONE, DOWN) or agent.detect(REDSTONE, RIGHT):
    agent.interact(LEFT)
```

Hint: "down **and** right" — both must be true. Move forward before you interact.

Write the fix:

Why was it wrong? Why does your fix work?

4 · Write a Branch ✍️

Add one more `elif` to the solver. When there is redstone straight **ahead**, the agent should turn **right** and move forward.

Write the branch:

In one line: where in the loop should this go, and why?

5 · Make It Your Own 🛠️

Open the extension world and run the solver with `run`. Change **one thing**, predict, then test.

Pick **one** (or your own):

- Add a `back` command that turns the agent around (two right turns).
- Change `agent.move(DOWN, 4)` to a new number and watch where it lands.
- Add an `elif` that reacts to redstone straight **ahead**.

Which change? Write the code.

Predict: what will happen?

Test it. What actually happened? If it surprised you, why?

6 · Show Your Work 📸 🎥

When the agent finishes the cube maze, come back.

Record **one video** (a phone is fine). Show two things:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

> Today I built ______.
>
> I built it using this code: ______.
>
> In this code I used ______.
>
> The hardest part was ______.
>
> That part was hard because ______.

Write your lines here, then say them in your video.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.